

28	To observe diffraction rings by a carbon film, a beam of electrons is accelerated from rest across a potential difference of V so that the de Broglie wavelength of the electrons is 0.10 nm. What is the value of V ?							
	A	90 V	B	150 V	C	270 V	D	330 V

Answer: B

de Broglie wavelength, $\lambda = \frac{h}{p}$

$$p = \frac{h}{\lambda}$$

$$\text{KE} = \frac{1}{2}mv^2 = \frac{p^2}{2m} = \frac{\left(\frac{h}{\lambda}\right)^2}{2m} = \frac{h^2}{2m\lambda^2}$$

Loss in EPE = Gain in KE

$$eV = \frac{h^2}{2m\lambda^2}$$

$$(1.60 \times 10^{-19})V = \frac{(6.63 \times 10^{-34})^2}{2(9.11 \times 10^{-31})(0.10 \times 10^{-9})^2}$$

$$V = 150.79 = 150 \text{ V (2 sig. fig.)}$$

20 At time t , a sample of a radioactive substance contains N atoms of a particular nuclide. At time